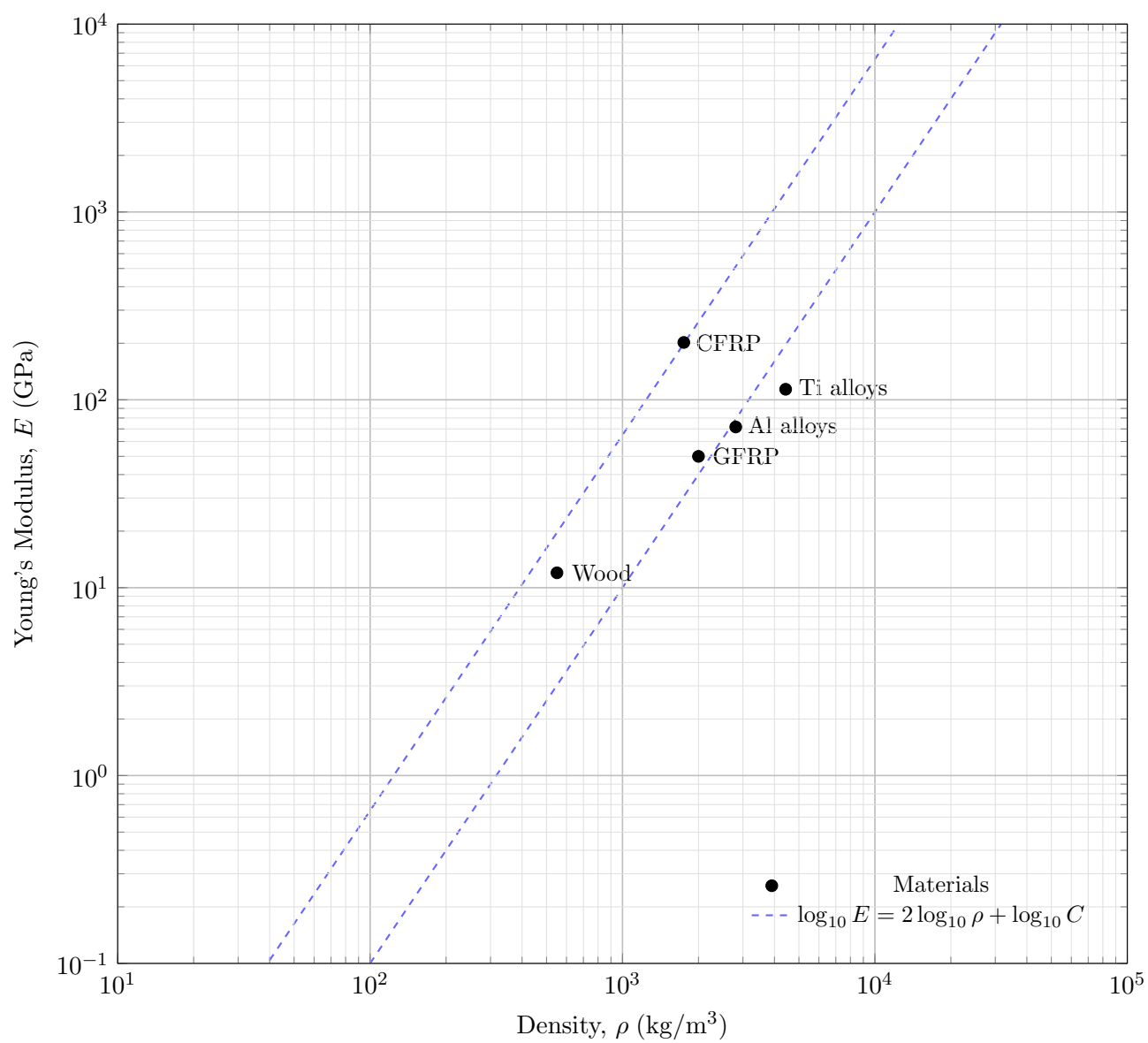




Objective (Maximize): Bending stiffness EI

$$\text{Material index } M = \frac{E}{\rho},$$

$$\log E = \log \rho + \log M,$$

Cost model

The unit manufacturing cost for the blow-moulded CFRP part is written as

$$C = C_{\text{mat}} + C_{\text{ded}} + C_{\text{cap}} + C_{\text{oh}} + C_{\text{nde}}. \quad (1)$$

Each contribution is

$$C_{\text{mat}} = C_m \frac{m}{1-f}, \quad \text{material cost}, \quad (2)$$

$$C_{\text{ded}} = \frac{C_t}{n} \left\lceil \frac{n}{n_t} \right\rceil, \quad \text{dedicated tooling cost}, \quad (3)$$

$$C_{\text{cap}} = \frac{L_f C_c}{t_{wo} \dot{n}}, \quad \text{capital cost}, \quad (4)$$

$$C_{\text{oh}} = \frac{\dot{C}_{oh}}{\dot{n}}, \quad \text{overhead cost}, \quad (5)$$

$$C_{\text{nde}} = \frac{C_{c,\text{NDE}}}{n} + C_{\text{coup}}, \quad \text{ultrasonic NDE cost per part}. \quad (6)$$

Here

- C_m : material price (USD/kg),
- m : component mass (kg),
- f : material waste fraction ($1-f$ is the utilisation),
- C_t : tooling cost (USD),
- n : batch size (number of parts),
- n_t : tool life (parts per tool),
- L_f : load factor,
- C_c : machine capital cost (USD),

- t_{wo} : capital write-off time (years),
- $\dot{n}$: production rate (parts/hr),
- $\dot{C}_{oh}$: overhead rate (USD/hr),
- $C_{c,NDE}$: capital cost of the ultrasonic NDE equipment (USD),
- C_{coup} : ultrasonic couplant cost per part (USD/part).

Table 1: Batch Size Parameters
Material Cost = 10 USD/kg Component Mass = 1 kg Overhead Rate = 110 USD/hr
Time = 5 yrs Load Factor = 0.5

Parameter	Low	High	Average
Capital cost (USD)	4.92×10^3	4.92×10^4	2.46×10^4
Material utilization fraction	0.5	0.8	0.65
Production rate (units/hr)	30	300	165
Tooling cost (USD)	4.92×10^3	1.64×10^4	1.065×10^4
Tool life (units)	1×10^5	1×10^7	5.05×10^6

Taking into account Non-Destructive Evaluation (NDE)

Estimated Ultrasonic Test equipment cost: **USD 3700**

Estimated couplant cost with a rate of 2 mL per part: **0.03 USD/part**

Cost of NDE per part (for a batch size $n = 2500$) is given by

$$C_{nde} = \frac{C_{c,NDE}}{n} + C_{coup} = \frac{3700}{2500} + 0.03 = 1.51 \text{ USD/part.}$$

The manufacturing cost per part without NDE is

$$C_{base} = 35.22 \text{ USD/part.}$$

Including NDE, the total cost per part is

$$C_{total} = C_{base} + C_{nde} = 35.22 + 1.51 = \mathbf{36.73 \text{ USD/part.}}$$

Cost model as a function of batch size

The unit cost C as a function of batch size n is

$$C(n) = C_{\text{mat}} + C_{\text{ded}}(n) + C_{\text{cap}} + C_{\text{oh}} + C_{\text{nde}}(n). \quad (7)$$

With the CES-type model the individual terms are

$$C_{\text{mat}} = C_m \frac{m}{1-f}, \quad (8)$$

$$C_{\text{ded}}(n) = \frac{C_t}{n} \left\lceil \frac{n}{n_t} \right\rceil, \quad (9)$$

$$C_{\text{cap}} = \frac{L_f C_c}{\dot{n} t_{wo}}, \quad (10)$$

$$C_{\text{oh}} = \frac{\dot{C}_{oh}}{\dot{n}}, \quad (11)$$

$$C_{\text{nde}}(n) = \frac{C_{c,\text{NDE}}}{n} + C_{\text{coup}}. \quad (12)$$

Using the average values from Table 1 and the NDE assumptions

$$\begin{aligned} C_m &= 10 \text{ USD/kg}, & m &= 1 \text{ kg}, & 1-f &= 0.65, \\ C_t &= 1.065 \times 10^4 \text{ USD}, & n_t &= 5.05 \times 10^6, \\ C_c &= 2.46 \times 10^4 \text{ USD}, & L_f &= 0.5, \\ \dot{n} &= 165 \text{ parts/hr}, & t_{wo} &= 5 \text{ yr}, \\ \dot{C}_{oh} &= 110 \text{ USD/hr}, \\ C_{c,\text{NDE}} &= 3700 \text{ USD}, & C_{\text{coup}} &= 0.03 \text{ USD/part}, \end{aligned}$$

the cost function used to generate the *Cost vs Batch Size* plot is

$$C(n) = 10 \frac{1}{0.65} + \frac{1.065 \times 10^4}{n} \left\lceil \frac{n}{5.05 \times 10^6} \right\rceil + \frac{0.5(2.46 \times 10^4)}{165 \times 5} + \frac{110}{165} + \frac{3700}{n} + 0.03. \quad (13)$$

Numerically, the constant part of (13) is

$$C_{\text{const}} = 10 \frac{1}{0.65} + \frac{0.5(2.46 \times 10^4)}{165 \times 5} + \frac{110}{165} + 0.03 \approx 30.99 \text{ USD/part},$$

so that

$$C(n) \approx 30.99 + \frac{1.065 \times 10^4}{n} \left\lceil \frac{n}{5.05 \times 10^6} \right\rceil + \frac{3700}{n}.$$

Base manufacturing cost C_{base} for $n = 2500$

The base cost (excluding NDE) is

$$C_{\text{base}}(n) = C_{\text{mat}} + C_{\text{ded}}(n) + C_{\text{cap}} + C_{\text{oh}}. \quad (14)$$

For a batch size $n = 2500$,

$$C_{\text{mat}} = 10 \frac{1}{0.65} \approx 15.38 \text{ USD/part}, \quad (15)$$

$$C_{\text{ded}}(2500) = \frac{1.065 \times 10^4}{2500} \left\lceil \frac{2500}{5.05 \times 10^6} \right\rceil = \frac{1.065 \times 10^4}{2500} \approx 4.26 \text{ USD/part}, \quad (16)$$

$$C_{\text{cap}} = \frac{0.5 (2.46 \times 10^4)}{165 \times 5} \approx 14.91 \text{ USD/part}, \quad (17)$$

$$C_{\text{oh}} = \frac{110}{165} \approx 0.67 \text{ USD/part}. \quad (18)$$

Thus

$$C_{\text{base}}(2500) = 15.38 + 4.26 + 14.91 + 0.67 \quad (19)$$

$$\approx \mathbf{35.22} \text{ USD/part}. \quad (20)$$

Including NDE for $n = 2500$,

$$C_{\text{nde}}(2500) = \frac{3700}{2500} + 0.03 \approx 1.51 \text{ USD/part},$$

and the total cost is

$$C(2500) = C_{\text{base}}(2500) + C_{\text{nde}}(2500) \approx 35.22 + 1.51 = \mathbf{36.73} \text{ USD/part}.$$